

Feature Normalisation

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Feature Normalisation: [0,1]-Scaling

- ▶ There are various ways to normalise (scale) a numerical features into a common scale.
- ▶ Method: [0-1] Scaling
$$\hat{x} = \frac{x - \min(x)}{\max(x) - \min(x)}$$
where the minimum and maximum values of the feature are computed over all training data points.
- ▶ The values of the scaled feature will now be in the interval [0-1].

Feature normalisation: Gaussian Normalisation

- ▶ Method 2: Gaussian Normalisation

$$\hat{x} = \frac{x - \mu}{\sigma}$$

- ▶ where the mean (μ) and the standard deviation (σ) for the feature are computed over all training data points. After this transformation each feature will have a zero mean and a unit variance.
- ▶ It is “easier” to compare two features, ignoring their absolute scales.